$\left[mid = \frac{s+e}{2} \right] \times$ binary search

$-ve$

$m+1$

$\boxed{\text{Sorted}}$: 1, 2, 3, ... $2^{31}-1$

s 0 e

$n = max$

int + max

no mid

$\frac{m+1}{2}$

max e

$\frac{s+e}{2}$ $-ve$ arr[-]

LC TLE

$$m = s + \frac{e-s}{2} \quad \Big| \quad \frac{s+e}{2} \times$$

avoid boundary condition

$$= \frac{2s + e - s}{2}$$

$$= \frac{s+e}{2}$$

OOB

Arrays: → An array is a fixed size data structure that stores elements of the same data type in contiguous memory locations.

(Homogeneous Data)

int

2	9	8	6	4
4b	4b	4b	4b	4b
1000	1004	1008	1012	1016

⇒ Contiguous memory

data = [1, "str", False];

Heterogeneous Data

Python (List)

char

a	b	c	d	f
1B	1B	1B	1B	1B
1000	1001	1002	1003	1004

* Key points:

- → Fixed Size
- → Index – based (starts with 0)
- → Homogeneous data
- → Stored in heap memory (for objects)
- → Faster access using index

Default Values in Arrays: (in C → we get garbage)

Data Type	Default Value
int	0
double	0.0
boolean	false
char	'\u000'
Object	null

{ Array elements are automatically initialized to their default values in Java. }

→ (UTF-8)